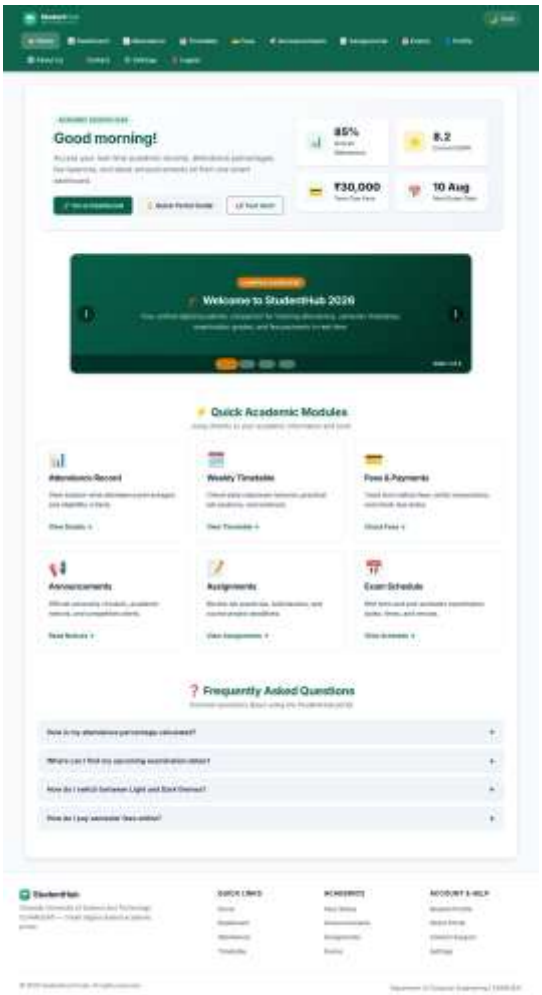
PRACTICAL-15

AIM: Integrate all modules of the StudentHub portal, test the complete system, finish the documentation and deploy it locally (Apache virtual host).

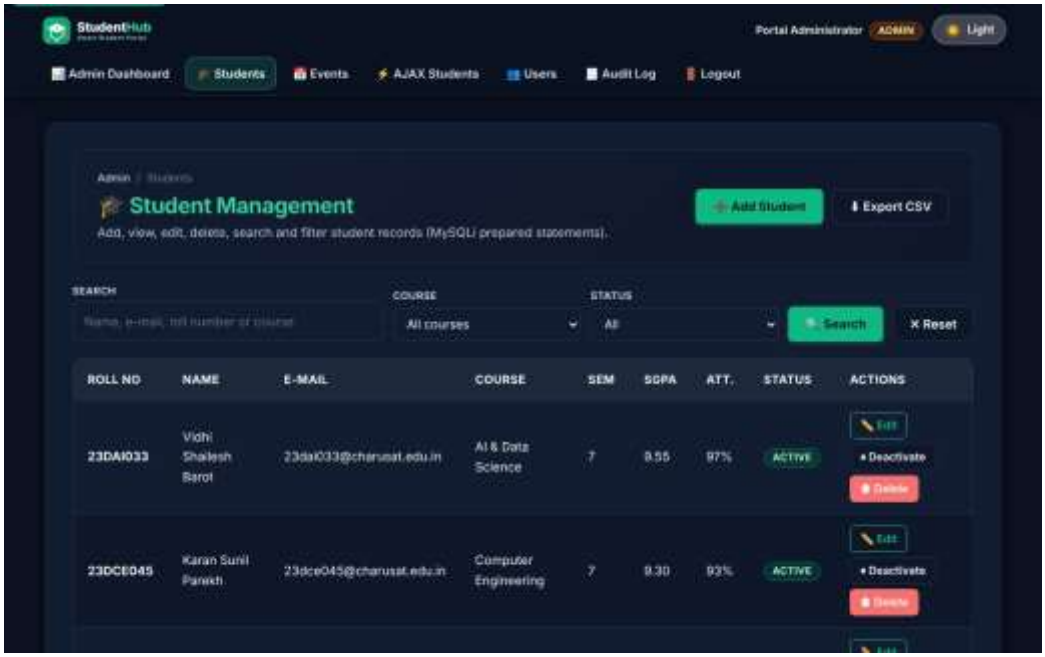
SOURCE CODE :

https://github.com/rachhadiyabrind-del/WDF_B/tree/main/Practical(1-4)(Studenthub)/Practical-15

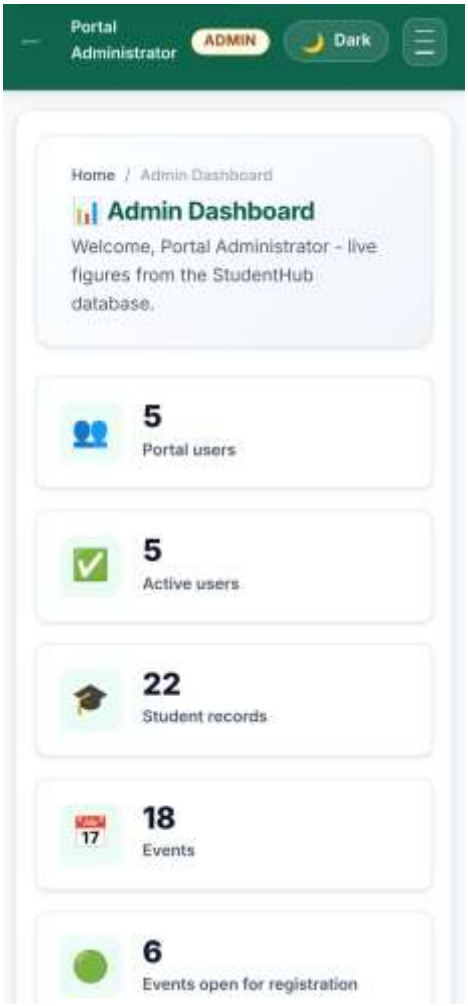
PREVIEW:



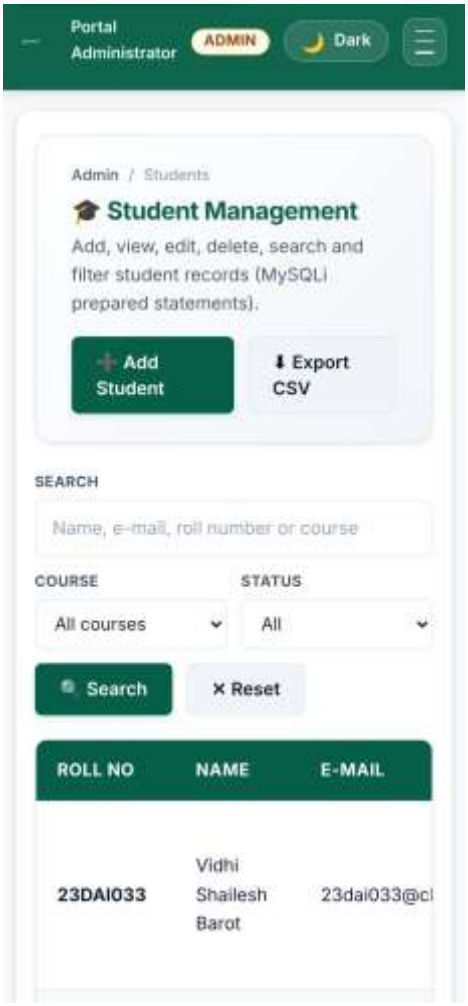
Home page - entry point of the portal



Dark theme on a PHP module page



Mobile layout of the admin dashboard (390 px)



Mobile layout of the student list

TEST CASES :

#	Input / action	Expected result	Result
1	P5 Validation	Inline errors, strength meter, CAPTCHA, aria-invalid	Pass
2	P6 JSON	Search, filter, sort, pagination, error state	Pass
3	P7 PHP form	POST validated; CSV + JSON files written	Pass
4	P8 Database	Schema, FK/UNIQUE/CHECK, PDO + MySQLi connection	Pass
5	P9 Registration	Hashed passwords, duplicates caught, CSRF	Pass
6	P10 Sessions	Login, logout, roles, timeout, lock-out, remember-me	Pass
7	P11 Students	CRUD, search, filter, CSV export, soft delete	Pass
8	P12 Events	CRUD, poster upload validation	Pass
9	P13 API	JSON endpoints, status codes, AJAX CRUD	Pass
10	P14 Admin	Dashboard, user toggle, audit log	Pass
11	Link check: crawl of 114 URLs as guest, student and admin	0 broken links (no 4xx/5xx)	Pass
12	PHP syntax check (php -l) on every .php file	No errors	Pass

KEY QUESTIONS :

1. Are all pages linked and navigable?

Yes. A crawler logged in as guest, student and admin and followed every link, image, stylesheet and script on the PHP pages and the HTML home page (114 URLs) - no response was 4xx or 5xx. Role-based menus link the modules to each other, and breadcrumbs show the position. The skip link, theme switch and hamburger menu are part of the shared header.

2. Do sessions, roles, validations and database operations work end to end?

Yes. The flow register (P9) -> login (P10) -> student dashboard, and login as admin -> dashboard (P14) -> students (P11) / events (P12) / AJAX (P13) / users / audit log was tested with scripted requests and with a real browser (Playwright); the screenshots in Practicals 5-14 were taken from that run. All write operations leave an audit_logs entry.

3. Are the security and accessibility checklists completed?

See the two tables below.

4. Is the documentation sufficient for installation and testing?

Yes. README.md in the project root explains requirements, installation (copy to htdocs, run Practical-15/reset_db.sh or import studenthub_full.sql, set DB credentials), demo logins, the folder structure, how to run each practical and how to test; Practical-15/TEST_REPORT.md lists every test case. Practical-15/studenthub.conf is a ready Apache virtual host.

5. Is the repository clean and properly committed?

Each practical is a separate folder so it can be committed separately with a clear message (for example "Practical 9: secure registration with password_hash"). The included .gitignore keeps logs, local settings and temporary files out of the repository; the repository contains only source, SQL, documents and screenshots. No real passwords or personal data are included - the seed data are demo records and the demo passwords are stored as bcrypt hashes.

SECURITY CHECKLIST :

Item	Done	Where
Prepared statements for every query	Yes	all PHP modules (MySQLi; PDO demo in P8)
Passwords hashed with password_hash / verified with password_verify	Yes	P9 register.php, P10 login.php
Output escaped with htmlspecialchars (e())	Yes	includes/helpers.php
CSRF token on every POST form and on API writes (header)	Yes	helpers.php, api/bootstrap.php
Session cookie HttpOnly + SameSite, strict mode, ID regenerated on login	Yes	helpers.php, auth.php

Idle session timeout	Yes	auth_boot()
Role check on the server for every admin page / endpoint	Yes	require_role(), api_require_admin()
Brute-force slow-down on login (5 attempts / 30 s)	Yes	P10 login.php
Same error for unknown user and wrong password	Yes	P10 login.php
Upload: size, real MIME, getimagesize, random name, execution blocked	Yes	includes/upload.php, uploads/.htaccess
Remember-me token stored only as SHA-256 hash, rotated on use	Yes	auth.php, remember_tokens
Generic error messages; details only in the error log	Yes	db.php, api/bootstrap.php
CSV export protected against formula injection	Yes	P11 students_export.php
Audit log of important actions	Yes	audit_log(), P14
HTTPS and production DB user with limited rights	To do on a real server	README (deployment notes)

ACCESSIBILITY CHECKLIST :

Item	Done	Where
Semantic landmarks (header, nav, main, footer) and a skip link	Yes	layout.php, HTML pages
One h1 per page, headings in order	Yes	all pages
Every form control has a label	Yes	all forms
Errors announced with role="alert" and aria-invalid	Yes	forms, validation.js
Status messages use role="status" / aria-live	Yes	AJAX page, JSON pages
Current page marked with aria-current	Yes	menu
Images have alt text (posters: "Poster of ...")	Yes	P12 pages
Full keyboard operation, visible focus	Yes	all pages
Dialog is a native <dialog> (focus trap, Esc to close)	Yes	P13
Light/dark themes with sufficient contrast	Yes	common.css
Responsive layout down to 390 px	Yes	screenshots above

DEPLOYMENT :

Local deployment: copy the project to the web root (XAMPP htdocs or /var/www/studenthub), import Practical-15/studenthub_full.sql (or run Practical-15/reset_db.sh), check the credentials in Practical-8/db.php and open Practical-10/login.php. Intermediate extension: Practical-15/studenthub.conf is an Apache virtual host (studenthub.local) with directory-listing off, SQL/shell files denied and security headers. Advanced extension: the static part (Html folder, for_css, javascript, Practical-6 JSON) works without PHP and can be published on GitHub Pages; the PHP modules need a PHP host.